

When Losing Is Not an Option: Competitive Priming and Support for January

6

Ryan DeTamble, David C. Barker, Marc J. Hetherington and Allen Wilson

Abstract: Observational research has linked sympathy for the January 6 insurrection to hypercompetitiveness, an understudied dispositional urge to win by any means necessary. Doubt remains, however, about 1) the causal effect of hypercompetitive impulses on electoral democracy and 2) the features of competitiveness more broadly. In this investigation, we demonstrate the causal influence of hypercompetitiveness by randomly assigning a subset of non-Democrats to an apolitical competitive game. Exposure appears to trigger more hypercompetitiveness, which in turn predicts January 6 sympathy. Moreover, as theorized, hypercompetitiveness' mediated effect is pronounced among those with high levels but absent among those with low levels. As a conceptual validity check, we further demonstrate that a different type of competitiveness that is not associated with a willingness to break rules has no influence. These results enhance causal leverage in support of hypercompetitiveness as a driver of January 6 support, and they invite more scholarship on the politics of personal competitiveness.

On January 6, 2021, the United States faced a crisis long assumed to be confined to fragile democracies: a violent attempt to prevent the peaceful transfer of presidential power. Although the insurrectionists themselves only numbered in the tens of thousands, polling has consistently found that a substantial share of Republicans and some Independents—numbering in the tens of millions—continue to characterize the day’s events favorably (Dress 2022; Orth 2023; Orth and Frankovic 2023). As the attack was directed at blocking the certification of an electoral defeat, such positive assessments seem to indicate the abdication of Loser’s Consent, the foundational democratic norm that electoral losers accept the winners’ authority until the next contest (Anderson et al. 2005; Dahl 1989; Esaiasson 2011; Bowler 2016).

Most scholars have explained this historic refusal to concede as a reflection of (a) partisan tribalism arising from affective polarization (Iyengar et al. 2019; Kalmoe and Mason 2022), (b) misinformation and conspiratorial thinking (Longwell 2022; Armaly, Buckley, and Enders 2022), and/or (c) authoritarian deference to Donald Trump (Perlstein 2021). More novelly, Hetherington et al. (2026) offer a theoretical alternative rooted in a psychological disposition toward *hypercompetitiveness* (HC), which is an antisocial need to win at all costs (Horney 1937; Ryckman et al. 1990; Orosz et al. 2018). Unlike other highly competitive

people who are motivated to achieve rather than to dominate—known in the extant literature as *personal development competitiveness* (PDC)—hypercompetitors care little about the integrity of competition; they are disproportionately willing to lie, bully, and cheat in pursuit of victory (Mudrack et al. 2012; Menesini et al. 2018; Orosz et al. 2018). Because the events of January 6 can be interpreted as precisely such a strategy, Hetherington et al. argue that support for the insurrection—and sympathy for it—may naturally proceed from such hypercompetitive impulses. While the authors’ panel data offer robust empirical support for their theory, such data alone cannot not inspire confident causal claims.

In this investigation, we apply a sterner test of hypercompetitiveness’s causal impact via a pre-registered experiment. By randomly exposing some non-Democratic survey respondents to a competitive priming task—a completely apolitical but competitive mental-rotation speed game—we aimed to trigger any latent competitive tendencies that the survey respondents possessed before asking them to evaluate the events of January 6. We find that exposure to such a competitive environment does indeed appear to activate hypercompetitiveness among a theoretically expected subset of respondents, which in turn is associated with more sympathetic evaluations of the insurrection. The results, however, do not suggest that being competitive, in general, is in any sense democratically

problematic; as we expected, personal development competitiveness bore no relation to January 6 evaluations.

Taken together, the results suggest that it is specifically hypercompetitiveness that may predispose some citizens to excuse or endorse anti-democratic behavior, and that this predisposition can be experimentally activated rather than merely inferred from observational correlations. These findings also suggest that individual competitiveness in its multiple forms may be a profoundly important political determinant more generally. In the next section, we flesh out our theoretical framework before turning to our experimental design and results.

Types of Competitiveness and Sympathy for the January 6 Riot

Emerging victorious in a free and fair election is only the first piece of the political puzzle of the peaceful transfer of power in democracies. Losers must also acknowledge their defeats and recognize the winners' authority. When electoral losers fail to concede and/or try to usurp power from the winners, it imperils the democratic system. Because support for January 6 suggests sympathy for a violent effort intended to reverse an electoral defeat, it serves as a uniquely valid indicator of public support for the refusal to grant Loser's Consent (Pape and Ruby 2021; Layman et al. 2023; Hetherington et al. 2026). Discovering and understanding the

psychological determinants of this refusal is of both scholarly and practical urgency.

The concept of hypercompetitiveness (HC) has not received much attention in political science, but it has deep roots in social psychology. First described as a neurotic drive to dominate and prevail in all domains of life (Horney 1937), the concept has been distilled over the years to capture an indiscriminate compulsion to conquer—by any means. While almost no one *prefers* to lose, to the hypercompetitive, victory is not just a desirable end state; it is a psychological necessity. Because those who score high in hypercompetitiveness tend to have fragile self-esteem (Ryckman et al. 1990), winning is “the only thing” that matters (Sanders 1948). Victory fuels self-worth and suppresses feelings of inadequacy, while losing poses an existential threat to the ego (Horney 1937; Ryckman et al. 1990; Newby and Klein 2014). The combination produces an ends-justify-the-means logic that is consistent with a willingness to endorse norm-violating or even violent tactics—both of which occurred on January 6—in pursuit of victory (Terpstra et al. 1993; Mudrack et al. 2012; Menesini et al. 2018).

Not all who emphatically embrace competition meet the conceptual definition of a hypercompetitor, however. In fact, social psychologists identify a far more commonly held competitive urge in the population, labeling it personal development competitiveness (PDC). In contrast with HC, PDC is a prosocial

disposition in which competition serves as an avenue for self-improvement (Ryckman et al. 1996; Ryckman et al. 1997; Hibbard and Buhrmester 2010). Even though competition is a fundamental feature of politics, political scientists have also overlooked this orientation as a potential political determinant. Unlike hypercompetitors, personal development competitors value competition as a means for growth and excellence, not as a pursuit of dominance. They respect the competitive process and their fellow competitors. Hence, they are not disposed to break the rules to gain an advantage.¹

Hetherington et al. (2026) use observational survey data to demonstrate a strong association between HC (both contemporaneous and lagged) and January 6 sympathy. However, their study is limited in a few key respects, which we list here in ascending order of importance. First, they employ an unconventional measure of hypercompetitiveness rather than any that have been repeatedly used and validated in the social psychology literature for years (see especially Orosz et al. 2018), which raises questions of measurement validity.

¹ To illustrate the difference between the two types of competitiveness, imagine a game of Monopoly. Were they on the edge of defeat, hypercompetitors would be likely to upend the Monopoly board to avoid losing—but personal development competitors would not. Likewise, personal development competitors would not be inclined to deny an electoral loss, but, again, hypercompetitors would.

Second, Hetherington et al. do not account for PDC, which is both normatively and empirically relevant to understanding January 6 support. Because survey reports of PDC tends to be moderately correlated with those of HC (very few people report high HC but low PDC [Newby and Klein 2014; Menesini et al. 2018; Orosz 2018]), it is conceivable that some portion of the HC effect that Hetherington et al. observe is in fact attributable to PDC; if so, the normative implications of the finding would change entirely, suggesting that any type of individual competitiveness is democratically dangerous. Theoretically speaking, we do not anticipate such a finding, but empirical testing is necessary to rule it out.

Third, Hetherington et al. do not examine the intersection of HC as a dispositional *trait* and as a situational *state*. Many inherent psychological tendencies do not automatically present themselves and exert influence over individual behavior; they often require situational activation or encouragement (e.g., Riker 1986, Iyengar and Kinder 1987, Barker 2002; Mendelberg 2001). Indeed, Reese et al. (2002) have shown that highly competitive frames can induce greater competitive desire than do less competitive frames. At the same time, no amount of activation matters if the latent proclivity does not already exist to at least some degree; in fact, much as certain persuasive arguments or presentations of evidence may sometimes induce backlash and a “hardening” of perspectives among

audiences who are dispositionally resistant to them (e.g., Bail et al. 2018; Nyhan and Reifler 2010 [but see Wood and Porter 2019; Guess and Coppock 2020]; Redlawsk 2002; Taber and Lodge 2006), noncompetitive individuals may recoil and retreat in response to competitive primes (Ryckman et al. 2009; AUTHORS REDACTED 2026).

Finally, and most importantly, the data on which Hetherington et al. rely are constrained in terms of their ability to confer causal leverage. Their panel data is useful in establishing temporal order—supporting the intuition that January 6 positivity does not cause HC—but it cannot rule out the possibility that some other combination of variables inspire both high levels of HC and positive appraisals of January 6. A more compelling research design would evaluate the extent to which the situational activation of latent HC corresponds to increased sympathy for the insurrection, thereby eliminating alternative explanations for the relationship. Such an approach would extend a long tradition of experimental research in political science and communication studies that use exogenous treatments to trigger latent dispositions, boosting self-reports of those dispositions and producing attitudinal changes (e.g., Albertson and Gadarian 2015; Banks and Valentino 2012; Iyengar and Kinder 1987; Landau et al. 2004; Oxley et al. 2008; Transue 2007; Valentino, Hutchings and White 2002).

Hypotheses

This theoretical framework leads us to put forward the following hypotheses, which are themselves related. The first is regarding the total effect of competitive priming on attitudes toward the insurrection:

H₁: *Exposure to an apolitical competitive context boosts sympathy for the January 6 insurgency.*

The second hypothesis—the “mediation hypothesis”—concerns the mechanism through which H₁ operates:

H₂ (mediation): *The treatment triggers latent HC, boosting it in survey responses, which in turn engenders more January 6 sympathy.*

The third hypothesis—the “moderation” hypothesis—likewise posits that HC conditions the treatment effect, but in a different way:

H₃ (moderation): *Empirical support for H₁ magnifies as HC grows larger.*

To test these hypotheses, we designed a novel survey experiment in which non-Democratic respondents were randomly assigned to play an apolitical mental-rotation speed game—which we framed as a competition against an opponent. In theory, respondents who possessed at least some degree of HC and/or PDC stood a chance of being “activated” by the treatment, which we examined in relation to their subsequently reported judgments of January 6 events. As an apolitical treatment, it represents a conservative test of our hypotheses.

Experimental Design, Data and Measures

In early January, 2024, using the Psytoolkit platform (Stoet 2010, 2017), we preregistered and fielded an online survey experiment with the Bovitz/Forthright survey panel ($n = 811$).² Although it is not probability-based, the Forthright panel provides a respondent pool that is more diverse and more representative of the U.S. voting-age population than are many alternative opt-in panels such as Lucid

² We only pre-registered the central hypothesis for this project, that “respondents randomly assigned to the hypercompetitive speed game task will have more sympathetic attitudes toward the Capitol insurrection of January 6th than the control group.” Thus, our other hypotheses (H2 and H3), while not preregistered, are the theoretical mechanisms that underpin our theory; we established them in that theory from the beginning, before empirically testing the data, and they represent a more stringent test of the theory.

or Amazon Mechanical Turk. All respondents completed the survey on a personal computer and passed an in-survey attention check.

After providing informed consent and confirming that they were at least 18 years old, participants answered a brief battery of political questions that included party identification and 2020 presidential vote choice (in the general election). Unbeknownst to them, respondents were then randomly assigned to one of two conditions. ***Treatment group*** respondents received the following prompt: “Now, you get to play in a brief ‘matching’ speed game, competing against an opponent who’s trying to beat you.” ***Control group*** respondents, by contrast, were simply instructed to “Please click ‘next’ to continue.”³

As shown in Figures 1 and 2, the mental rotational gameplay task presented treatment group respondents with a two-dimensional object at the top of their computer screens and asked them to choose as quickly as possible which of two bottom-of-screen objects—both rotated—matched the one at the top. The game included five practice rounds of moderate difficulty followed by ten scored rounds of higher difficulty. Upon completion, players were shown their accuracy rate and

³ The treatment and control groups did not differ meaningfully on demographics with one exception: Nonwhite respondents were slightly more likely to be randomly assigned to the control group. Because they also tended to score slightly higher on HC in our sample, we estimated alternative prediction models in which we included Nonwhite identity as a covariate. Doing so did not affect the results at all.

average response time per round.⁴ The entire activity, including the training rounds and the scored rounds, took most participants about one minute to complete.

Figure 1. Gameplay Instructions

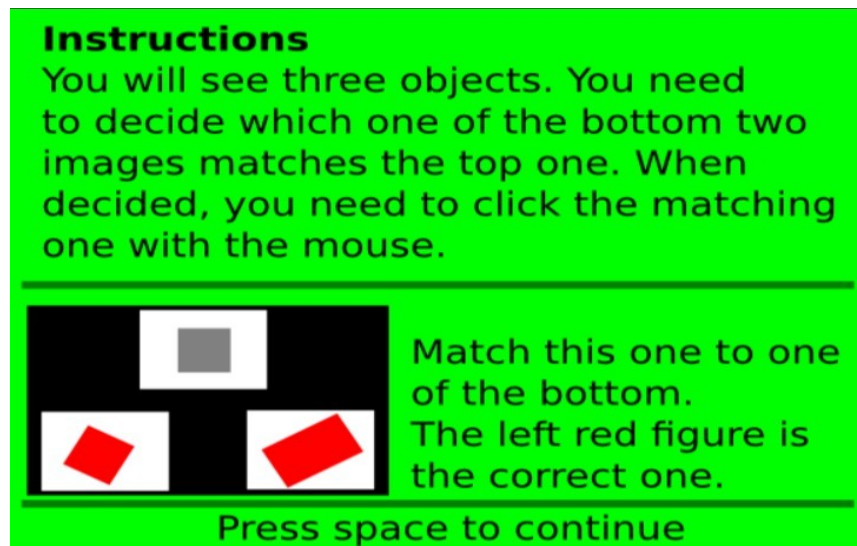
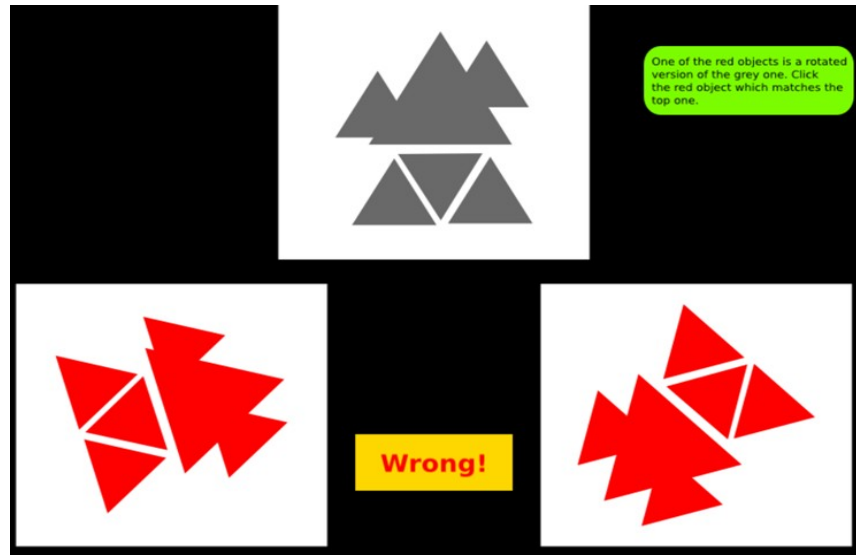


Figure 2. Sample Gameplay Screen⁵

⁴ This particular version of the mental rotation game was utilized partially because it was not exceedingly easy or overly onerous. All respondents managed to answer at least two of the ten rounds correctly. In fact, all but 11 respondents managed to answer at least four of the 10 rounds correctly. However, only 8 respondents managed to get every single round correct. The modal gameplay result was seven correct 7 answers correct and 3 answers wrong.

⁵ This is an example of the screen during a gameplay situation. The orange “Wrong!” box pops on the screen following incorrect object selection. Alternatively, when the user selects the right object, a green “Correct!” box pops up on the screen.



We selected this particular task for five reasons: (1) it has a well-established history in social-psychology experiments (e.g., Shepard and Metzler 1971; Vandenburg and Kuse 1978; Voyer et al. 1995); (2) it is not a popular game or sport about which participants may hold prior attitudes or known skill levels; (3) it places participants under time pressure, which is known to heighten competitive arousal (Malhotra 2010); (4) it is apolitical; and (5) it is well suited to online survey administration.

We consider the game a “weak” treatment, because (a) it was very brief, (b) it was administered online, through a screen, to survey respondents who were not necessarily motivated to be fully engaged, and (c) the purported opponent is invisible. Therefore, the experiment’s “intent to treat” surely did not translate into

an actual treatment for many respondents.⁶ Perhaps even more consequential with respect to this treat vs. intent to treat dilemma, it is likely not possible for the treatment to induce an increase in hypercompetitiveness among people who score very low in hypercompetitiveness by disposition. Approximately one-third of the US population is fundamentally noncompetitive and finds competitive environments distasteful (AUTHORS REDACTED 2026). Because hypercompetitiveness, specifically, is an antisocial disposition, moderate and high levels are relatively rare in the population. As such, those very low in HC are very unlikely to have been moved by the treatment to feel more hypercompetitive. In fact, the treatment may have prompted some of them to feel even less so—and perhaps even less sympathetic toward January 6 events—than they otherwise would have been. Ideally, we would have been able to measure HC at one point in time and delete respondents from the sample who revealed no penchant for HC whatsoever, before executing the experiment at a later point in time. Because we

⁶ Even so, additional analyses show that respondents' accuracy levels within the competitive game differed greatly by HC and PDC levels. Moving from 0% accuracy to 100% accuracy corresponded to a 27-percentage-point drop in HC ($p = 0.01$) but no statistical difference in PDC ($p = 0.68$). We suspect this association can be attributed to respondents seeing the word "Wrong!" in bold, red letters when they answered incorrectly, demoralizing high-HC respondents (who tie their feelings of self-worth to winning) but not others (including high PDC respondents). While only suggestive, we take this as additional evidence that the experimental prime worked as intended, and that there are substantive psychological differences between hypercompetitors and personal development competitors.

were unable to do that, their presence in the sample places substantial analytical pressure toward null results in relation to our first hypothesis.

Measuring January 6 Sympathy

Next, the survey asked all respondents to consider the following prompt: “On January 6, 2021, supporters of then President Trump overran the Capitol building where Vice President Pence and members of the U.S. Senate were counting electoral votes from the 2020 presidential election. Choose the word or words from each pair that you feel best describes what happened.”⁷

(1) Acceptable vs. Unacceptable

(2) Violent Insurrection vs. Legitimate Political Discourse

(3) Justified vs. Unjustified

⁷ Respondents were also asked if they felt that the events of January 6 were “patriotic” or “unpatriotic” and if they thought they were “planned” or “unplanned.” Though these items were preregistered, we exclude them from this analysis because they may not best capture the concept of unwillingness to grant loser’s consent. For instance, one can believe that the events of January 6 were planned and still support them; likewise, one can believe that they were “unplanned” but still oppose them. Similarly, if one expresses that the events of January 6 were “patriotic” it is likely that they have bought into the idea that the election was rigged and that their partisan team did not actually lose the race. Such items are more indicative of conspiratorial thinking rather than support for the January 6 insurrection.

We coded each sympathetic response (“acceptable,” “legitimate political discourse,” and “justified”) as “one” and the corresponding unsympathetic response as “zero.” We next estimated a one-factor confirmatory factor analysis using maximum-likelihood estimation to generate respondent level factor scores, which we arranged on a 0 to 1 continuum, creating an index of *January 6 Sympathy*. All three items load strongly on a single factor (Eigenvalue = 1.71; factor loadings: .74, .64, .87, respectively, $\alpha = .79$).⁸

Measuring Hypercompetitiveness and Personal Development Competitiveness

To measure hypercompetitiveness, we depart from Hetherington et al. (2026)—in keeping with the competitiveness canon in social psychology—by using Orosz et al.’s (2018) Hypercompetitive Orientation battery.⁹ Respondents indicated their agreement-disagreement on seven-point scales with three statements: (1) “The most important thing is winning, no matter what”; (2) “I am willing to do whatever it takes to win”; (3) “I will do anything to win, even nasty things.” A generalized maximum-likelihood factor analysis confirmed that all three items

⁸ The roughly three-year lag between the Capitol insurrection and our survey fielding stood to potentially attenuate any possible association between experimentally induced competitiveness and sympathy toward those events, thereby further biasing results toward the null.

⁹ Ryckman and his colleagues (1990, 1994, 1997) devised a measure for the disposition, which was later refined by Orosz et al. (2018).

load strongly on a single factor (Eigenvalue = 1.98; factor loadings: .86, .81, and .75, respectively; $\alpha = .85$). Again, following convention in the social psychology literature (Newby and Klein 2014; Orosz et al. 2018; Menesini et al. 2018), we saved that factor score (rescaled to 0-1) as our base measure.

To measure personal development competitiveness, we asked respondents to indicate their agreement-disagreement to Orosz et al.'s (2018) self development competitiveness battery statements on a 7-point Likert scale (rescaled to 0–1 for analysis): (1) “Competitive situations allow me to bring the best out of myself”; (2) “I enjoy testing myself in competitive situations”; (3) “I enjoy competition as it allows me to discover my abilities.” A generalized maximum-likelihood confirmatory factor analysis confirmed that all three items load strongly on a single factor (Eigenvalue = 2.27; factor loadings: .87, .84, and .90, respectively; $\alpha = .90$). We array the respondent factor scores for our Personal Development Competitiveness index on a 0 to 1 scale.

The frequency distributions of the two types of competitiveness differ fundamentally. The modal score for hypercompetitiveness, an *anti-social* desire to win no matter what, is the absolute minimum of the scale. Indeed, only about a quarter of respondents score even as high as the midpoint or above. Some might be tempted to view its distribution as evidence of a design flaw in the survey items, but it is not. In a highly functioning society, we would expect to find vanishingly

few individuals who strongly agree that they would “do anything to win, even nasty things.” However, readers probably can identify at least a few times in their lives when they have encountered people who would do just that. Very high levels of the disposition, then, ought to be rare. In contrast, personal development competitiveness is a *pro-social* disposition that can promote achievement. Given that the United States stands out among nations for its emphasis on the value of the competitive spirit, we should not be surprised to see a distribution with the lion’s share of the cases above the scale’s midpoint.

Analytic Models and Results

Because it makes little sense to think that hypercompetitiveness would induce sympathy among Democrats regarding an election that they won for an insurrection to overturn the results of that election sparked by Donald Trump, we restricted our analytic sample to Republicans, Republican-leaning Independents, and “pure” Independents who report no partisan lean.¹⁰ We also excluded respondents who completed the survey in fewer than five minutes because they

¹⁰ This is another (albeit minor) way in which our analysis departs from Hetherington et al. (2026), who did not include “pure” Independents in their sample, and our preregistration, which suggests including respondents of all partisan or non-partisan stripes. We maintain Independents here for two reasons: (1) to preserve statistical power for our randomized experiment (137 respondents were a lot to lose, particularly given H2 and H3), and (2) because some pure Independents were supporters of President Trump and therefore fell into the category of electoral “losers,” who were our focus—historically, very small percentages of Democrats reported supporting him.

would not have had sufficient time to have engaged in any meaningful way with either the experimental task or the rest of the survey items (the mean survey completion time was just over 29 minutes). Finally, we deleted respondents who took more than two hours to complete the survey (presumably taking breaks) because they would likely not have maintained any heightened state of hypercompetitiveness that the treatment might have induced by the time they answered the January 6 questions. The resulting analytic sample consists of 489 respondents (227 in the treatment group; 262 in the control group).

1. Simple Experimental Model and Results

First, to get a simple answer to our basic question—whether priming competitiveness causes greater sympathy for the January 6 insurrection—we began by estimating a linear regression model predicting the latter with the former. To address the “intent to treat” vs. “treat” dilemma we discussed earlier—the fact that, per our theory, a competitive treatment prime could not have induced January 6 sympathies among non-competitive or nearly non-competitive respondents—we estimated the model three times: first with the entire sample; then, after excluding respondents who reported zero levels of HC; and finally, after excluding respondents who reported levels that were lower than .2 on the 0-1 scale.

As Table 1 shows, the randomized treatment boosts January 6 sympathy in all three models, and, as expected, the coefficient estimate for the treatment creeps upward as we restrict the sample to those with more hypercompetitive dispositions. The treatment effect is about 7 percentage points in the model with no exclusions, increases to a little over 8 points after eliminating those scoring at the minimum of HC, and jumps to about 11 points after we eliminate all those who scored below .2 on HC. Decreasing the number of cases, of course, increases the size of the standard errors on the estimates, so the p-values for all three estimates are about the same for all three coefficients, ranging from $p=.06$, two tailed to $p=.03$, two-tailed.

Table 1:

	No Exclusions	Excluding HC=0	Excluding HC<.2
	Parameter Estimate (Standard Error) p-value	Parameter Estimate (Standard Error) p-value	Parameter Estimate (Standard Error) p-value
Constant	0.321 (0.026)	0.322 (0.028)	0.317 (0.035)
Treatment	0.070 (0.037) $p=.06$	0.082 (0.041) $p=.05$	0.113 (0.052) $p=0.03$
Number of cases	489	390	261

2. Hypercompetitiveness Mediation Models and Results

Having induced the necessary change in January 6 sympathy to carry out a causal test, we turn to exploring whether the effect of the competitive game treatment on January 6 sympathy worked through increasing hypercompetitiveness. The next analyses estimate the extent to which treatment exposure corresponded to higher levels of reported HC which, in turn, boosted *January 6 sympathy*. To test our hypothesis, we employed the causal mediation framework developed by Imai, Keele, and Tingley (2010), which provides a formal identification strategy for decomposing a total treatment effect into direct and indirect (mediated) components under clearly stated assumptions. Implementation requires two regression equations: one that uses *Hypercompetitiveness* as the outcome variable and the *Competitive Treatment* as the explanatory variable (using the probit estimator), and a second that uses *January 6 Sympathy* as the outcome variable and both the *Competitive Treatment* and *Hypercompetitiveness* as the explanatory variables (using the ordinary least squares estimator).¹¹

Recall again that a competitive game treatment is unlikely to induce an increase in hypercompetitiveness among those scoring at the low end of the scale.

¹¹ We obtained the confidence intervals for the regression coefficients using quasi-Bayesian Monte Carlo simulation, with 2,000 draws (Imai, Keele, and Tingley 2010).

To address this “intent to treat” challenge regarding non-hypercompetitive, we again conducted our analyses using three versions of the sample as above: an unrestricted version, a second version that removed respondents who scored “zero” in *HC*, and a third version that removed respondents who scored lower than .2 (on the 0-1 scale) in *HC*. A final issue to consider is what functional form the effect of hypercompetitiveness ought to take. Our theorizing throughout has suggested that linearity is unlikely, but the correct form is not obvious from theory. To gain some leverage of the question, we estimated the average treatment effect for respondents in 11 separate bands of hypercompetitiveness ($HC=0$, $.001 < HC < .10$; $.101 < HC < .20$; $.201 < HC < .30$, etc.). As expected, the treatment effects at the low end of *HC* are trivial. For the five bands below 0.40, the median treatment effect is 0.020. But, for the six bands above 0.4 *HC*, the median treatment effect is 0.110. Hence we simplified the measure of *HC* for our mediation model by dichotomizing it such that those scoring above .4 on the 0-1 scale are coded as “one” (29% of the effective sample) and those scoring below .4 are coded as “zero.”¹²

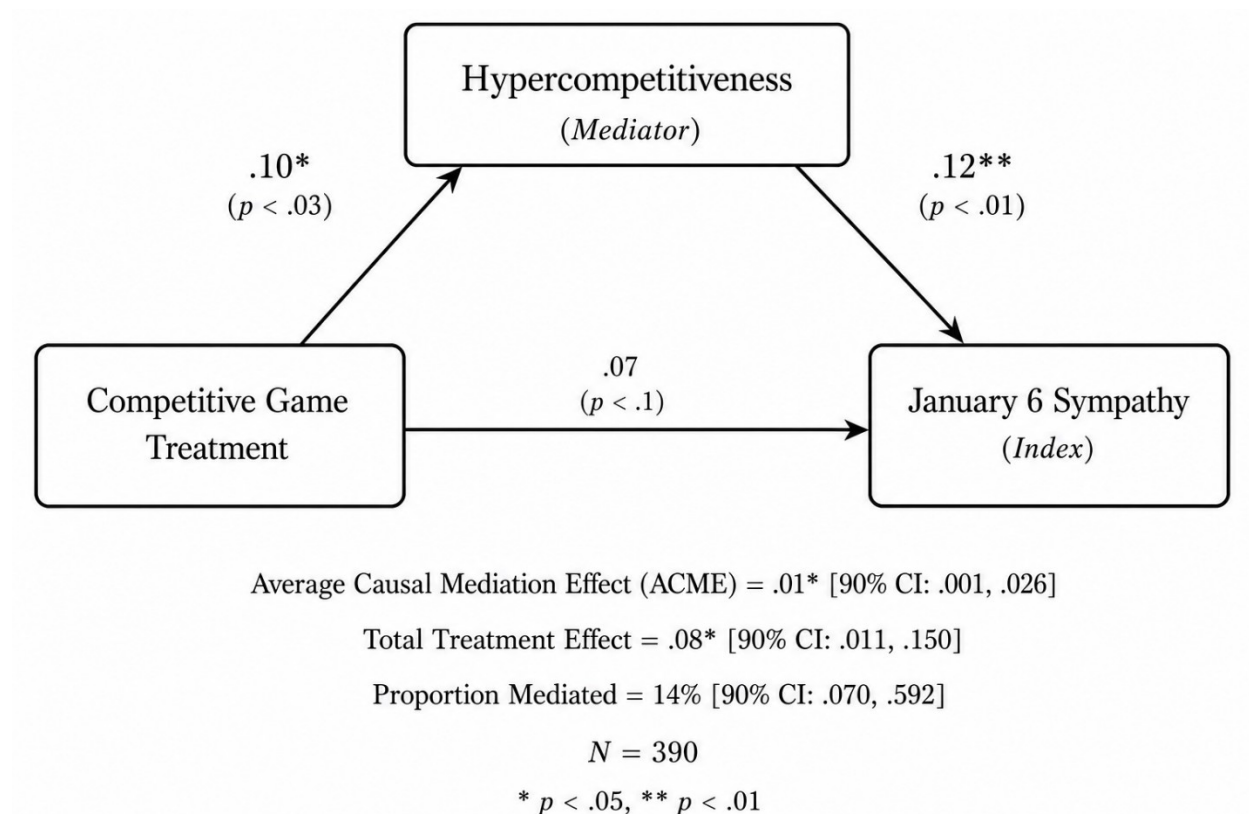
¹² Alternative model estimates use the full range of *HC* as the measure. As we show in the supplementary materials online, those models reveal similar (though as we expected, somewhat weaker) results.

We created three figures, one for the full sample, one for respondents who scored above the minimum in *HC*, and the last for those who scored .2 or above in *HC*. We present the second of the formulations as Figure 1, while the first and the third appear in the supporting information. The data maintain their consistency with the theory in these more stringent tests.¹³ In the unrestricted sample (SI section x), treatment exposure corresponded to an eight percentage-point increase in *HC* ($p < .05$), which in turn corresponded to a twelve percentage-point increase in *January 6 Sympathy* ($p < .01$). That translated into an average mediated treatment effect (ACME) of .01 ($p < .05$) and a total treatment effect of .07 (12% mediated; $p < .05$). The results strengthened only slightly in the first restricted model (Figure 1): treatment exposure corresponded to a ten percentage-point increase in *HC* ($p < .03$), which corresponded again to a twelve percentage-point increase in *January 6 Sympathy* ($p < .01$) and that translated into an average mediated treatment effect (ACME) of .01 ($p < .05$) as well as a total treatment effect of .08 (14% mediated; $p < .05$). Meanwhile, as we predicted, the results strengthened substantially in the second restricted model (SI section x): treatment exposure corresponded to an 18 percentage-point increase in *HC*

¹³ An additional discriminant validity check reveals that by contrast, the treatment did not succeed in priming the benign variant of competitiveness—personal development competitiveness—eliminating its prospect of supplying the causal pathway to heightened *January 6 Sympathy* (see the Supplementary Materials Online).

($p < .03$), which corresponded to a 14 percentage-point increase in *January 6 Sympathy* ($p < .01$) and that translated into an average mediated treatment effect of .01 ($p < .05$) and a total treatment effect of .11 (22% mediated; $p < .05$). In all three models, the non-mediated treatment effect (the direct correspondence between *Treatment Exposure* and *January 6 Support*, independent of *HC*) was statistically indistinguishable from zero (applying a lenient test at $p < .10$).

Figure 1:



These results support both Hypothesis 1 and Hypothesis 2: exposure to an apolitical hypercompetitive context significantly increased sympathy for the

January 6 insurgency among non-Democratic respondents, and it did so by triggering increases in hypercompetitiveness, particularly among those scoring at the high end of the scale.

Personal Development Competitiveness Mediation Models and Results

Although we believe that both hypercompetitiveness and personal development competitiveness are valuable concepts for understanding how individuals approach politics, they are conceptually different, and, hence, ought to affect different variables. Empirically speaking, many people are highly competitive (they score high in PDC) without believing that breaking the rules to win is appropriate (they don't score high in HC). Only those who score high in hypercompetitiveness ought to be inclined to be political rule breakers and norm busters. When dependent variables reflect support for ignoring rules and norms, as is the case in our study, only hypercompetitiveness should affect beliefs. Demonstrating that a distinction exists between the two concepts is useful to helping social scientists apply them.

To test whether the theoretical distinction stands up to empirical scrutiny, we replicate the analysis we carried out above for personal development competitiveness. As with the prior analysis we begin with no restrictions on how

high people score in PDC (SI section y), then we exclude respondents with zero PDC (Figure 2), and last exclude respondents with <0.2 PDC (SI section y).

As predicted, the results for PDC differ substantially from those for HC. None of the models for personal development competitiveness produces a mediated effect on January 6 support. Indeed, personal development competitiveness is not a significant predictor of January 6 support in either the absence or presence of a prime. In contrast, hypercompetitiveness was positively associated with January 6 support even without the prime, but the prime revealed its causal effect. Taken together, the results make clear that HC and PDC are fundamentally different concepts. In the PDC analysis, we observe a small but statistically meaningful direct effect of the competitive game treatment on January 6 support for two of the PDC samples—those that include all respondents and those that excluded only those scoring at the absolute minimum of PDC—but, importantly, the effect is *not* mediated through PDC. The results presented above suggest that hypercompetitiveness is likely the variable through which the effect was mediated. Regardless, we can rule out PDC as a cause for increased support for the insurrection.

3. Moderation Models and Results

Recalling that we theorized that *HC* would not only mediate the effect of *Treatment Exposure* on *January 6 Sympathy*—providing its causal mechanism—we also posited that it would do so nonmonotonically, with the slope of *Treatment Exposure* steepening as *HC* progresses up the scale. To evaluate this hypothesis (H_3), we estimated a final linear regression model predicting *January 6 Sympathy* that included an interaction term to the right-hand side of the equation that multiplied *Treatment Exposure* by *HC*, along with each constitutive variable.

As Figure 4 highlights, among those with low or even mean levels of *HC*, *Treatment Exposure* caused no significant increase (substantively or statistically) in *January 6 Sympathy* (.30 to .32 or .34, with overlapping CIs). Among those who reported the highest levels of *HC*, by contrast, *Treatment Exposure* corresponded to a 26 percentage-point spike in *January 6 Support* ($p < .001$). In comparison, a minimum-to-maximum increase in *HC* among control group respondents only corresponded to an eight percentage-point bump in *January 6 Sympathy*, which was itself statistically insignificant.

We assessed the robustness of these findings by re-estimating both all four of the OLS equations with a battery of control variables: race, gender, age, income, education, religiosity, ideological identification, authoritarianism, and anti-

intellectualism. The inclusion of these covariates—either jointly or in various combinations—did not alter the substantive patterns reported above. The direction, approximate magnitude, and statistical significance of the treatment effect on both hypercompetitiveness and sympathy for January 6 insurgents remained consistent across all specifications.

Conclusion

Electoral democracy survives only so long as losers concede their electoral losses. The January 6 insurrection—and the persistent sympathy it commands among a sizable share of Republican voters—represents a serious challenge to that foundational norm of American political life. Previous observational work established that hypercompetitiveness, the psychological need to win by any means, is strongly associated with sympathetic evaluations of the Capitol attack (Hetherington et al. 2026). In the present investigation, we attempted to move beyond correlational evidence to causal evidence—by randomly assigning respondents to a brief, apolitical competitive game before they evaluated the events of January 6. This is, to our knowledge, the first experimental evidence that competitive contexts can causally activate sympathy for anti-democratic action by triggering hypercompetitiveness.

The results are notable in their implications. Among non-Democratic respondents, a few minutes of competitive gameplay increased the probability of expressing sympathetic characterization of the insurrection. This effect emerged despite the fact that the competitive task had no political content whatsoever, that respondents were unaware they were being primed, and that the events being evaluated had occurred approximately three years before the survey was fielded. Each of these design features biases *against* finding an effect, making the observed results all the more noteworthy.

The causal mediation analysis further illuminates the psychological pathway. Approximately 18-23% of the total treatment effect is transmitted through heightened self-reported hypercompetitiveness, as measured by Orosz et al.'s (2018) Hypercompetitive Orientation battery. This indirect effect is statistically significant and substantively meaningful, demonstrating that a competitive experience can measurably shift individuals' conscious endorsement of a "win at all costs" mentality. This shift, in turn, predicts greater sympathy for the insurrection.

Moreover, our evidence suggests that a different typology of competitiveness, personal development competitiveness, has no direct, mediated, or moderated effect on January 6 support. This finding clarifies that different dispositions toward competition differ in their impact. In other words, motivation

matters: Those who are driven to competition in order to better themselves are not the same as those who are driven to competition for the sole purpose of winning, degrading opponents and boosting their own feelings of self-worth.

These findings suggest implications that extend beyond feelings about January 6. If a brief, low-stakes competitive experience administered through a computer screen can increase positive evaluations of the most chaotic transition of presidential power in American history, it could be that the competitive political environments that pervade modern life (e.g. common media portrayals of electoral politics as sport come to mind) may be having similar effects on a far larger and more consequential scale, particularly as elections become increasingly high-stakes (e.g., Layman et al. 2023). The results suggest that democratic resilience is not only a function of citizens' fixed traits and civic education but may also be shaped by the situational context in which citizens form political judgments. Contexts that heighten the salience of competition may, perhaps inadvertently, erode the willingness to accept electoral losses that feel existential—to one's political goals, yes, but to one's ego and feelings of self-worth as well.

It could be argued that the defining feature of modern representative democracy is competition via elections. Interestingly, while political scientists have studied those systems of competition intensely, they have largely ignored the role that individual-level competitiveness and competitive arousal may play in

motivating all sorts of political actors—from voters, to donors, to candidates. Our finding that primed hypercompetitiveness can impact political attitudes that strike at the heart of the democratic system itself may also serve as a first step towards a future of researchers uncovering the competitive origins of all sorts of political preferences. If an apolitical speed shape-selection game can prime feelings of hypercompetitiveness, competitive political frames with higher perceived stakes against political rivals are likely do so to an even higher extent, from there influencing changes in political attitudes and actions.

Several limitations warrant acknowledgment. First, the competitive prime was delivered through a screen rather than in person with a visible opponent, which likely attenuates its psychological intensity. That we nonetheless observed significant effects speaks to the robustness of the underlying mechanism, and laboratory or field experiments with more immersive and engaging competitive tasks might reveal even larger effects. Second, the roughly three-year gap between the events of January 6 and the survey fielding may have weakened any association between experimentally induced competitiveness and attitudes that had, for many respondents, already crystallized. Future studies fielded closer in time to contested political events could capture the mechanism while the content is most potent. Third, while our analytic sample of non-Democratic respondents is theoretically appropriate, it limits our ability to generalize to the full electorate.

Whether competitive priming might activate sympathy for norm-violating political action among Democrats or Independents in other contexts remains an open question.

Finally, the present study complements aforementioned earlier observational findings in an important way. Hetherington et al. (2026) demonstrated that hypercompetitiveness, measured as a stable personality disposition, predicts insurrectionist sympathy even when measured months before the Capitol attack occurred. The present study demonstrates that the same psychological infrastructure can be *situationally activated* by environmental competition. Taken together, the two studies suggest that the relationship between competitiveness and anti-democratic sentiment operates at both the dispositional and the contextual level—a dual pathway with troubling implications for democratic health in societies where political competition is increasingly zero-sum in character. Understanding and addressing the competitive roots of democratic erosion should be a priority for scholars and practitioners alike.

References

- Albertson, Bethany and Shana Kushner Gadarian. 2015. *Anxious Politics: Democratic Citizenship in a Threatening World*. Cambridge University Press.
- Anderson, Christopher J., André Blais, Shaun Bowler, Todd Donovan, and Ola Listhaug (eds). 2005. *Losers' Consent: Elections and Democratic Legitimacy*. Oxford University Press.
- Armaly, Miles T., David T. Buckley, and Adam M. Enders. 2022. "Christian Nationalism and Political Violence: Victimhood, Racial Identity, Conspiracy, and Support for the Capitol Attacks." *Political Behavior*, 44: 937–960.
- Bail, Christopher A., Lisa P. Argyle, Taylor W. Brown, John P. Bumpus, Haohan Chen, M.B. Fallin Hunzaker, Jaemin Lee, Marcus Mann, Friedolin Merhout, and Alexander Volfovsky. 2018. "Exposure to opposing views on social media can increase political polarization." *Proceedings of the National Academy of Sciences*, 115(37): 9216-9221.
- Barker, David C. 2002. *Rushed to Judgment: Talk Radio, Persuasion, and American Political Behavior*. Columbia University Press.

Bowler, Shaun. 2016. "I study democracies, and what I've learned is this: they collapse without graceful losers." Vox. October 14th, 2016. URL: <https://www.vox.com/the-big-idea/2016/10/14/13277626/losers-democratic-transition-sanders-trump>

Brown, Steven P., William L. Cron, John W. Slocum Jr. 1998. "Effects of Trait Competitiveness and Perceived Intraorganizational Competition on Salesperson Goal Setting and Performance." *Journal of Marketing*, 1998, 62(4): 88-98.

Carvalho, Catarina L., Isabel R. Pinto, Rui Costa-Lopes, Darío Paéz, and José M. Marques. 2021. "Support for group-based inequality among members of low-status groups as an ingroup status-enhancement strategy." *Social Psychological Bulletin*, 16(2): 1-27.

Dahl, Robert A. 1989. *Democracy and Its Critics*. New Haven, CT: Yale University Press

DiMenichi, Brynne C., and Elizabeth Tricomi. 2015. "The power of competition: Effects of social motivation on attention, sustained physical effort, and learning." *Frontiers in Psychology*, 6: 145588.

Dress, Brad. 2022. "New poll finds 33 percent of GOP support actions of Jan. 6 rioters."

The Hill, June 16, 2022. URL: <https://thehill.com/homenews/state-watch/3526469-new-poll-finds-33-percent-of-gop-support-actions-of-jan-6-rioters/>

Duckitt, John. 2001. "A Dual Process Cognitive-Motivational Theory of Ideology and Prejudice." *Advances in Experimental and Social Psychology*, 33.

Duckitt, John and Chris G. Sibley. 2010. "Personality, Ideology, Prejudice, and Politics: A Dual-Process Motivational Model." *Journal of Personality*, 78:6.

Esaiasson, Peter. 2011. "Electoral losers revisited—How citizens react to defeat at the ballot box." *Electoral Studies*, 30(1): 102-113.

Fletcher, Thomas D., Debra A. Major, and Donald D. Davis. 2008. "The Interactive Relationship of Competitive Climate and Trait Competitiveness with Workplace Attitudes, Stress, and Performance." *Journal of Organizational Behavior*, 29(7): 899-922.

Gneezy, Uri, Muriel Niederle, and Aldo Rustichini. 2003. "Performance in Competitive Environments: Gender Differences." *The Quarterly Journal of Economics*, 118(3): 1049-1074.

- Guess, Andrew and Alexander Coppock. 2020. "Does Counter-Attitudinal Information Cause Backlash? Results from Three Large Survey Experiments." *British Journal of Political Science*, 50(4): 1497-1515.
- Guimond, Serge, Michael Dambrun, Nicolas Michinov, and Sandra Duarte. 2003. "Does social dominance generate prejudice? Integrating individual and contextual determinants of intergroup cognitions." *Journal of Personality and Social Psychology*, 84(4): 697.
- Hetherington, Marc J., Allen Wilson, Ryan M. DeTamble, and David C. Barker. 2026. "Hypercompetitiveness and Loser's Consent." *Public Opinion Quarterly*, 90(1): 202–217.
- Hibbard, David R. and Duane Buhrmester. 2010. "Competitiveness, Gender, and Adjustment Among Adolescents." *Sex Roles*, 63: 412–424.
- Horney, Karen. 1937. *The Neurotic Personality of our Time*. W&W Norton & Co.
- Huang, Li-Li and James H. Liu. 2005. "Personality and social structural implications of the situational priming of social dominance orientation." *Personality and Individual Differences*, 38: 267–276.

Imai, Kosuke, Luke Keele, and Dustin Tingley. 2010. "A general approach to causal mediation analysis." *Psychological Methods*, 15(4), 309–334.

Iyengar, Shanto and Donald R. Kinder. 1987. *News That Matters: Television and American Opinion*. University of Chicago Press.

Iyengar, Shanto, Yphtach Lelkes, Matthew Levendusky, Neil Malhotra, Sean J. Westwood. 2019. "The Origins and Consequences of Affective Polarization in the United States." *Annual Review of Political Science*, 22(1): 129-146.

Kalmoe, Nathan P. and Lilliana Mason. 2022. *Radical American Partisanship: Mapping Violent Hostility, Its Causes, and the Consequences for Democracy*. University of Chicago Press.

Landau, Mark J., Sheldon Solomon, Jeff Greenberg, Florette Cohen, Tom Pyszczynski, Jamie Arndt, Claude H. Miller, Daniel M. Ogilvie, and Alison Cook. 2004. "Deliver us from evil: The effects of mortality salience and reminders of 9/11 on support for President George W. Bush." *Personality and Social Psychology Bulletin* 30(9): 1136-1150.

- Layman, Geoffrey, Frances Lee, and Christina Wolbrecht. 2023. "Political Parties and Loser's Consent in American Politics." *The ANNALS of the American Academy of Political and Social Science*, 708: 164-83.
- Longwell, Sarah. 2022. "Trump Supporters Explain Why They Believe the Big Lie." *The Atlantic*. April 18th, 2022. URL:
<https://www.theatlantic.com/ideas/archive/2022/04/trump-voters-big-lie-stolen-election/629572/>
- Levin, Shana. 2004. "Perceived group status differences and the effects of gender, ethnicity, and religion on social dominance orientation." *Political Psychology*, 25: 31–48.
- Malhotra, Deepak. 2010. "The desire to win: The effects of competitive arousal on motivation and behavior." *Organizational Behavior and Human Decision Processes*, 111(2): 139-146.
- Menesini, Ersilia, Fulvio Tassi, and Annalaura Nocentini. 2018. "The competitive attitude scale (CAS): A multidimensional measure of competitiveness in adolescence." *Journal of Psychol. Clin. Psychiatry* 9(3): 240-244.

- Mendelberg, Tali. 2001. *The Race Card: Campaign Strategy, Implicit Messages, and the Norm of Equality*. Princeton University Press.
- Morrison, Kimberly R., Nathanael J. Fast, and Oscar Ybarra. 2009. "Group status, perceptions of threat, and support for social inequality." *Journal of Experimental Social Psychology*, 45: 204–210.
- Morrison, Kimberly R. and Oscar Ybarra. 2008. "The effects of realistic threat and group identification on social dominance orientation." *Journal of Experimental Social Psychology*, 44: 156–163.
- Mudrack, Peter E., James M. Bloodgood, and William H. Turnley. 2012. "Some Ethical Implications of Individual Competitiveness." *Journal of Business Ethics*, 108: 347–359.
- Newby, Jennifer L. and Rupert G. Klein. 2014. "Competitiveness Reconceptualized: Psychometric Development of the Competitiveness Orientation Measure as a Unified Measure of Trait Competitiveness." *The Psychological Record*, 64: 879-895.
- Nyhan, Brendan and Jason Reifler. 2010. "When Corrections Fail: The Persistence of Political Misperceptions." *Political Behavior*, 32(2): 303-330.

Orosz, Gábor, Istvan Toth-Kiraly, Noemi Buki, Kristztian Ivaskevics, Beata Bothe, and

Marta Fulop. 2018. "The Four Faces of Competition: The Development of the Multidimensional Competitive Orientation Inventory." *Frontiers in Psychology*, 9(779): 1-16.

Orth, Taylor and Kathy Frankovic 2023. "A small but growing share of Americans approve of the people who took over the Capitol on January 6." *YouGov*. March 16, 2023. URL: https://today.yougov.com/politics/articles/45427-growing-share-americans-approve-jan-6-riot?redirect_from=%2Ftopics%2Fpolitics%2Farticles-reports%2F2023%2F03%2F16%2Fgrowing-share-americans-approve-jan-6-riot

Orth, Taylor. 2023. "Most Americans — but fewer than in 2021 — disapprove of the January 6 Capitol takeover." *YouGov*. January 4th, 2023. URL: https://today.yougov.com/politics/articles/44904-most-americans-disapprove-january-6-capitol-attack?redirect_from=%2Ftopics%2Fpolitics%2Farticles-reports%2F2023%2F01%2F04%2Fmost-americans-disapprove-january-6-capitol-attack

Oxley, Douglas R., Kevin B. Smith, John R. Alford, Matthew V. Hibbing, Jennifer L. Miller, Mario Scalora, Peter K. Hatemi, and John R. Hibbing. "Political Attitudes Vary with Physiological Traits." *Science*, 321(5896): 1667-1670.

Pape, Robert A. and Keven Ruby. 2021. "The Capitol Rioters Aren't Like Other Extremists:

We analyzed 193 people arrested in connection with the January 6 riot—and found a new kind of American radicalism." *The Atlantic*. February 2nd, 2021. URL: <https://www.theatlantic.com/ideas/archive/2021/02/the-capitol-rioters-arent-like-other-extremists/617895/>

Perlstein, Rick. 2021. "The Long Authoritarian History of the Capitol Riot." *New York*

Magazine: Intelligencer. June 30th, 2021. URL: <https://nymag.com/intelligencer/2021/06/long-authoritarian-history-january-6-insurrection.html>

Redlawsk, David P. 2002. "Hot Cognition or Cool Consideration? Testing the Effects of

Motivated Reasoning on Political Decision Making." *The Journal of Politics*, 64(4): 1021-1044.

Reese, Zachary A., Stephen M. Garcia, and Robin S. Edelstein. 2022. "More than a game:

Trait competitiveness predicts motivation in minimally competitive contexts." *Personality and Individual Differences*, 185: 111262.

Riker, William H. 1986. *The Art of Political Manipulation*. Yale University Press.

Ryckman, Richard. M., Cary R. Libby, Bart van den Borne, Joel A. Gold, and Marc A.

Lindner, M. 1997. "Values of Hypercompetitive and Personal Development Competitive Individuals." *Journal of Personality Assessment*, 69: 271-283.

Ryckman, Richard M., Max Hammer, Linda M. Kaczor, and Joel A. Gold. 1996.

"Construction of a Personal Development Competitive Attitude Scale." *Journal of Personality Assessment*, 66(2): 374-385.

Ryckman, Richard M., Max Hammer, Linda M. Kaczor, and Joel A. Gold. 1990.

"Construction of a Hypercompetitive Attitude Scale." *Journal of Personality Assessment*, 55: 630-639.

Ryckman, Richard M., Bill Thornton, J. Corey Butler. 1994. "Personality Correlates of the

Hypercompetitive Attitude Scale: Validity Tests of Horney's Theory of Neurosis." *Journal of Personality Assessment*, 62, 84-94.

Ryckman, Richard M., Bill Thornton, Joel A. Gold. 2009. "Assessing Competition

Avoidance as a Basic Personality Dimension." *The Journal of Psychology*, 143(2): 175-192.

Sanders, Red. 1948. As quoted in Ziffrin, Lester "UCLA Grid Team Takes Defeat Hard."

Los Angeles Times (October 18).

- Schmitt, Manfred, Nyla R. Branscombe, and Deborah M. Kappen. 2003. "Attitudes toward group-based inequality: Social dominance or social identity." *British Journal of Social Psychology*, 42: 161–186.
- Schnall, Simone, Jonathan Haidt, Gerald L. Clore, and Alexander H. Jordan. 2008. "Disgust as embodied moral judgment." *Personality and Social Psychology Bulletin* 34(8): 1096-1109.
- Schrock, Wyatt A., Douglas E. Hughes, Frank Q. Fu, Keith A. Richards, and Eli Jones. 2016. "Better Together: Trait Competitiveness and Competitive Psychological Climate as Antecedents of Salesperson Organizational Commitment and Sales Performance." *Marketing Letters*, 27(2): 351-360.
- Schurr, Amos, and Ilana Ritov. 2016. "Winning a competition predicts dishonest behavior." *Proceedings of the National Academy of Sciences*, 113(7): 1754-1759.
- Shepard, Roger N., and Jacqueline Metzler. 1971. "Mental rotation of three-dimensional objects." *Science*, 171(3972): 701–703.

- Sibley, Chris G., Marc S. Wilson, and John Duckitt. 2007. "Effects of Dangerous and Competitive Worldviews on Right-Wing Authoritarianism and Social Dominance Orientation over a Five-Month Period." *Political Psychology*, 28: 357-371.
- Sibley, Chris G., John Duckitt, Robin Bergh, et al. 2013. "A Dual Process Model of Attitudes towards Immigration: Person $\times$ Residential Area Effects in a National Sample." *Political Psychology* 34(4): 553–72.
- Stoet, Gijsbert. 2010. "PsyToolkit - A software package for programming psychological experiments using Linux." *Behavior Research Methods*, 42(4), 1096-1104.
- Stoet, Gijsbert. 2017. "PsyToolkit: A novel web-based method for running online questionnaires and reaction-time experiments." *Teaching of Psychology*, 44(1), 24-31.
- Taber, Charles S. and Milton Lodge. 2006. "Motivated Skepticism in the Evaluation of Political Beliefs." *American Journal of Political Science*, 50(3): 755-769.
- Terpstra, David E., Elizabeth J. Rozell, and Robert K. Robinson. 1993. "The influence of personality and demographic variables on ethical decisions related to insider trading." *The Journal of Psychology* 127(4): 375-389.

- Transue, John E. 2007. "Identity Salience, Identity Acceptance, and Racial Policy Attitudes: American National Identity as a Uniting Force." *American Journal of Political Science*, 51(1): 78-91.
- Valentino, Nicholas A., Vincent L. Hutchings, and Ismail K. White. 2002. "Cues That Matter: How Political Ads Prime Racial Attitudes During Campaigns." *American Political Science Review*, 96(1): 75-90.
- Vandenberg, Steven G. and Allan R. Kuse. 1978. "Mental rotations, a group test of three-dimensional spatial visualization." *Perceptual and Motor Skills*, 47(2), 599–604.
- Voyer, Daniel, Susan Voyer, and M.P. Bryden. 1995. "Magnitude of sex differences in spatial abilities: A meta-analysis and consideration of critical variables." *Psychological Bulletin*, 117(2), 250–270.
- Wood, Thomas and Ethan Porter. 2019. "The Elusive Backfire Effect: Mass Attitudes' Steadfast Factual Adherence." *Political Behavior*, 41(1): 135-163.
- Yip, Jeremy A., Maurice E. Schweitzer, and Samir Nurmohamed. 2018. "Trash-talking: Competitive incivility motivates rivalry, performance, and unethical behavior." *Organizational Behavior and Human Decision Processes*, 144: 125-144.